

Regression Utility of a 10,240-Byte Tabular Generator

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Abstract—The displayed DOPE generator is a compact residual: at most 12 screened inputs, a width-16 tanh hidden layer, and a refit linear readout. AdamW trains the hidden basis for 2,048 steps at learning rate 0.002, with no embedding. Each method keeps its own selection objective. After selection, artifacts are scored by retention, the share of real-versus-null validation improvement recovered by an auditor trained on synthetic rows, and by charged bytes. On 97 paired PMLB regression lineages the CatBoost median is 0.940 (95% lineage-bootstrap interval 0.914 to 0.976), against 0.656 for a native-selected Gaussian copula, 0.586 for a study Chow–Liu tree, and -0.012 for study independent marginals. Holm-adjusted Wilcoxon tests separate these three CatBoost comparisons. The CatBoost record against the copula is 86/0/11 wins/ties/losses. The linear auditor does not separate DOPE from the Gaussian copula (49/0/43). An unpooled 21-lineage block compares CTGAN and TVAE. An unpooled 6-lineage Forest-Flow block is ahead on CatBoost (1/0/5) at 115,202,253 to 1,085,155,062 bytes. 98/100 displayed DOPE artifacts sit inside the 10,240-byte cap. Official tests were not opened. Master Fitness v2 is null. TabSyn, a 100-lineage TabDDPM panel, the production privacy-attack panel, fit-seed variance, and the full ablation grid are not measured. Grouped-validation fidelity and an empirical distance ledger are reported. They are not differential privacy or HIPAA de-identification.

Index Terms—Synthetic tabular data, retention, artifact bytes, paired tests.

I. INTRODUCTION

A generator paper often reports the metric that also selected the configuration. Those native scores live on different scales: predictive efficacy for CTGAN and TVAE [1], held-out log density for a Gaussian copula [2], a forest density for ARF [3], and a diffusion or CART objective for other families [4], [5]. This paper separates selection from the common measurement. After each method has been selected on its own objective, the shared columns are retention and the bytes charged to the artifact.

The decision those columns can support is narrow. It is whether a network small enough to ship under a 10,240-byte cap still preserves downstream regression utility on real validation rows. It is not a declaration that one generator is best, and it is not a certification score. Master Fitness v2 (MFS-v2) is defined in Section III and is null on every method here, because the six components and the hard-gate map are not all present.

The contributions are:

- 1) The displayed residual and the sealed-test training pipeline (Section III, Fig. 1, (1)).
- 2) Retention, charged bytes, and grouped-validation fidelity, with lineage-bootstrap intervals (Fig. 2).

- 3) Paired Wilcoxon tests with Holm correction, rank-biserial correlations, and a Friedman–Nemenyi diagram on the common lineage set (Fig. 3).
- 4) A separate six-lineage Forest-Flow block, not pooled with the density block.
- 5) An explicit account of what this panel does not measure, including MFS-v2 (Section VIII).

An unexecuted method is not a win for DOPE. Density, neural, Forest-Flow, and BeyondArena blocks are not pooled.

II. RELATED WORK

Tabular synthesis has moved from parametric copulas [2] and Chow–Liu trees [6] through conditional GANs, including TableGAN, and variational autoencoders [1], [7] to diffusion and score-based models [4], [8]–[14]. Autoregressive and language-model generators [15], [16], prior-data fitted generators [17], [18], CTAB-GAN and CTAB-GAN+ [19], [20], and graphical neural generators [21] target the same mixed-type tables. ARF estimates density with a forest [3]. CART synthesis is the synthpop line [5]. A recent survey collects these design choices [22]. This paper runs the Gaussian copula, study Chow–Liu, study independent marginals, CTGAN, TVAE, and a 6-lineage Forest-Flow confirmation. TabSyn, TabDDPM on the 100-lineage panel, STaSy, CoDi, TabDiff, CTAB-GAN+, GReaT, REaLTabFormer, and TabPFGen are cited and not measured. SMOTE [23] was not run. This panel scores regression retention, not classification oversampling.

Differentially private Bayesian networks and marginal synthesizers [24]–[27] optimize a privacy budget. Membership inference and shadow-model attacks are a separate detection literature [28]–[30]. Formal differential privacy is a different claim from an empirical resemblance score [31], [32]. Utility reviews separate marginal fidelity from downstream prediction [33], [34]. Alpha-precision, beta-recall, SDMetrics, and synthcity are the usual fidelity reports [35]–[37]. The fidelity results add alpha-precision, beta-recall, a distance-to-closest-record share, and a DOMIAS AUC. ARF is absent because shared validation sampling is still pending. The production privacy-attack panel remains not measured. Nothing in those fields is a formal privacy guarantee or a HIPAA de-identification claim.

Tree models remain strong tabular predictors, which is why CatBoost is one of the three auditors [38]–[40]. TabZilla and TabArena show how brittle a single split can be [41], [42]. The regression lineages come from PMLB [43], [44]. Comparing several methods on the same lineages needs a rank test and a critical difference, or a Bayesian signed-rank alternative [45],

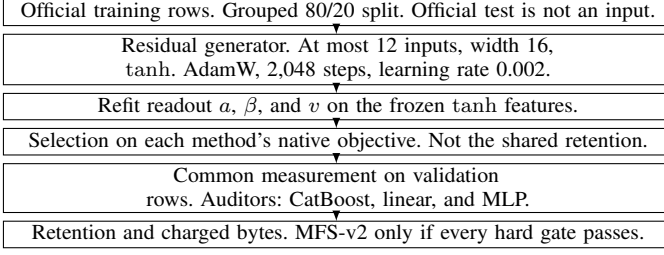


Fig. 1. Displayed pipeline. The official test is not an input.

[46]. This paper reports the frequentist procedure: a Friedman test and a Nemenyi critical difference on the common set, plus paired Wilcoxon tests. The Bayesian signed-rank procedure is not computed.

III. METHOD

A. Displayed generator

The displayed profile is `features12_steps2048`. It is a compact neural residual, not a GAN, a diffusion model, or a transformer. Fig. 1 is the pipeline. Training screens at most 12 inputs. A hidden layer of width 16 maps them through tanh. AdamW, learning rate 0.002, runs for 2,048 steps on the regression mean-squared error, with gradient clipping at 5. The hidden weights and biases are then frozen. The intercept, the linear skip, and the hidden readout are refit on those frozen features. Coefficients whose absolute value is below 10^{-5} are dropped from the skip. For a selected input vector x_S ,

$$h_j = \tanh(b_j + w_j^\top x_S), \quad \hat{y} = a + \beta^\top x_S + v^\top h. \quad (1)$$

Sampling adds the noise model stored in the artifact. There is no embedding table in (1). A separate dataset-action router uses a vector of dimension $856 + 24 \times (128 + 10) = 4,168$. That vector does not generate the rows in this paper.

The profile was fixed before the shared retention was inspected. Compile-time search records the proxy $0.45u + 0.25d + 0.30j$ and then keeps the candidate with the smallest normalized gate shortfall, then the smallest artifact, then the lowest candidate id. Every such compile is marked `production_evidence_unmeasured`. No global DOPE family has been selected. Other shapes exist and are not this result: micro-TVAEs, TinyMAT, joint TVAE, a masked transformer, and a TabSyn candidate. A direct-rank TabDDPM candidate is a research comparator with no claim on the 10,240-byte tier. The scored campaign did not store a per-step loss. A later logged replay did. The supplement plots that replay. The replay files are not substituted for the scored artifacts, and a lower hidden-basis loss at 8,192 steps does not replace the displayed profile. The full width-and-step grid is not measured.

B. Retention and MFS-v2

Let L_{null} , L_{real} , and $L_{\text{synthetic}}$ be the auditor's loss on real validation rows when that auditor is trained on a null baseline, on real fit rows, and on synthetic rows. Retention is

$$\frac{L_{\text{null}} - L_{\text{synthetic}}}{L_{\text{null}} - L_{\text{real}}}. \quad (2)$$

TABLE I
ILLUSTRATION OF (3). NOT A MEASUREMENT.

Gate map	Scalar
All six gates passed	82.40
Byte gate failed	null

A value of 1 means the synthetic-trained auditor matched the real-trained auditor. A value above 1 means it beat that auditor. A negative value means it did worse than the null. A lineage is informative when the real improvement is at least 0.01 times the absolute null loss and is not zero. Otherwise the lineage is omitted. It is not entered as zero. The three auditors are CatBoost [39], ordinary linear regression, and an MLP. The lineage score is the median of sample seeds 101, 211, and 307. The panel score is the median of those lineage scores. Intervals below are percentile lineage-bootstrap intervals, 10,000 resamples of lineages, with the three seeds kept inside the lineage median. They are not per-seed whiskers.

MFS-v2 is a different object. With component $c_i \in [0, 1]$, weight $w_i > 0$, $\epsilon = 10^{-6}$, and $W = \sum_i w_i$,

$$\text{MFS-v2} = 100 \exp\left(\frac{1}{W} \sum_i w_i \ln(\epsilon + c_i)\right) \quad (3)$$

when the hard-gate map is non-empty, every hard gate passed, and every positive-weight component is finite and inside $[0, 1]$. Otherwise the scalar is null. The weights are utility transfer 0.30, driver fidelity 0.20, distribution fidelity 0.20, structure fidelity 0.15, coverage realism 0.10, and compactness 0.05. A privacy failure removes the score through a gate. Utility transfer inside (3) is a clamped one-sided lower bound, not the median of (2). Compactness is $\text{clamp}(1 - \text{bytes}/10240, 0, 1)$. Crossing 10,240 bytes fails the byte gate, so the scalar stays null. An artifact over the cap is a failed gate, not a retention of zero.

Table I applies (3) to six invented components (0.90, 0.80, 0.85, 0.70, 0.75, 0.95). The resulting 82.40 is arithmetic, not a measurement. The second row fails the byte gate and the implementation returns null. No artifact in this paper has a measured MFS-v2 value.

IV. EXPERIMENTAL SETUP

A. Partitions and provenance

The panel is the rights-cleared PMLB regression collection [43], [44]. The data lock admits 104 MIT-licensed entries and prepares 100. 4 lineages have no split object because the same source-row digest appears in two partitions (`check_nonoverlap` raises on that overlap). Another 3,755 catalog entries lack a redistribution grant and are not backfilled. The prepared cut is a grouped 80/20 of the official training rows, seed 1729. Worker directories hold `train.csv` and `validation.csv`. They do not hold `test.csv`. Official tests stay sealed, and final evaluation is not authorized. Table II is the lock and the file-name count. Test-file contents were not read.

Fit seed 11 is the population actually measured. Sizes n and $4n$ are both stored. The tables below are $4n$. The Gaussian

TABLE II
TEST-SPLIT PROVENANCE. FILE-NAME COUNTS ONLY.

Item	Record
Catalog SHA-256	ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120
Eligible / prepared / overlap exclusions	104 / 100 / 4
Unknown-rights catalog entries held out	3755
Split	official_test_grouped_training_80_20, seed 1729
Overlap lineage ids	590210721973399a, 9fe58a656fdb458a, d06a584f07d0acd5, d0ba629ed07b5d63
Final evaluation authorized	false
S3 worker directories / train.csv / validation.csv	100 / 100 / 100
test.csv under S3 workers	0
BeyondArena workers / worker test.csv	5 / 0
Evaluator test.csv files, contents not read	5

copula row is Copulas 0.14.1 GaussianMultivariate, selected by held-out mean log density. CTGAN and TVAE are the author implementation, selected by the mean of linear and MLP regression R^2 on validation rows. Chow–Liu uses eight bins and Laplace smoothing 1.0 in study code. Independent marginals are a study reference. The ARF watch log’s last count is 799 closed and 799 ok out of 800 planned, then a sealing record. That log is not a retention panel, and ARF was not sampled on this schedule. Forest-Flow on 6 lineages uses the author regression objective on its frozen grid. Native objectives are not ranked across methods.

B. Statistics

Comparisons are paired on the lineages where both methods have a complete informative group. The tests are two-sided Wilcoxon signed-rank tests on the nonzero paired differences, and one-sided sign tests for a positive difference, with a matched-pairs rank-biserial correlation. Holm correction is applied inside a family, separately for each test, and is not pooled across families. On CatBoost the Holm sign-test values are 2.15×10^{-15} , 6.72×10^{-24} , and 4.95×10^{-27} . The density family has nine tests (three comparators times three auditors). The neural family is CTGAN and TVAE times the auditors. The Forest-Flow family is separate and has six lineages. A Friedman test and a Nemenyi critical difference use the four density methods on their common complete-informative set, with rank 1 assigned to the higher retention. The critical-difference constant for four methods at $\alpha = 0.05$ is 2.569. These are descriptive validation tests. They are not the production paired-superiority rule, which also requires a frozen evaluator and a gate-pass analysis. PTF-v1, MFS-v2, and release-safe L3 stay null.

C. Compute

Table III is receipt wall-clock and charged bytes. Fit and sample columns are `elapsed_seconds` when the receipt stores that timer. A timer, host, device, or byte field the receipt does not store is not recorded. CPU model is not recorded on every row. TabSyn is not measured. A full TabDDPM panel is not measured. The supplement states the replay walls, the Forest-Flow sums, and the unsplit density attempt timers.

V. RESULTS

Table IV keeps, for each block and auditor, the tuned baseline with the highest paired median retention. The difference is

DOPE minus that baseline. Its interval is the paired lineage bootstrap already used in the block tables. W/T/L counts lineages with a positive, zero, or negative difference. Holm p is the adjustment inside that block’s existing family. It is not a new family, and the blocks are not pooled. A positive median with a small p -value still includes lineage losses.

A. Density block

Fig. 2 plots retention against $\log_{10}$ charged bytes for the three auditors. The open markers are each method’s own median and lineage-bootstrap interval on the lineages that method completed. The table intervals are the paired subsets. For CatBoost those two DOPE sets coincide. Paired differences outside $[-1.5, 1.5]$ are counted in the panel title and omitted from that frame. The retention axis stops at 1.6. The dashed line is the cap. Neural and Forest-Flow artifacts are not on this figure.

Table V is the paired density block. Under CatBoost, DOPE’s median paired differences are positive against the Gaussian copula (0.224, interval 0.149 to 0.290), the study Chow–Liu tree, and the study independent marginals. The Holm-adjusted p -values are 1.35×10^{-13} , 4.85×10^{-16} , and 1.17×10^{-16} . The rank-biserial correlations are 0.890, 0.978, and 0.999. The Gaussian copula is the strongest density baseline. Wins, ties, and losses against it are 86/0/11 under CatBoost and 65/0/10 under the MLP. On the 97 lineages common to all four methods, the Friedman p -value is 2.71×10^{-44} and the critical difference is 0.476. All six CatBoost pairs exceed that difference. Fig. 3 shows the sorted paired differences. The bar is the critical difference.

The linear auditor does not separate DOPE from the Gaussian copula (49/0/43; Holm 0.373; difference 0.00366, interval -0.00438 to 0.012; rank-biserial 0.107). The interval contains zero. Ranks are 1.533 and 1.598 against critical difference 0.489. Chow–Liu and the independent marginals remain separated under the linear auditor and the MLP. A CatBoost separation is not a separation under every auditor.

Table VI is the median raw loss on the same informative lineages. RMSE is the square root of the stored mean squared error, and R^2 is one minus that loss divided by the training-mean null. Retention is the median of the per-lineage ratios, not the ratio of these medians. CatBoost reads RMSE 0.053 and 0.081, with R^2 0.881 and 0.792.

Table VII counts artifacts at or under the cap. The two displayed DOPE artifacts over the cap are 4544_Geographical

TABLE III
COMPUTE AND ARTIFACT COST. MISSING RECEIPT FIELDS ARE NOT RECORDED.

Method	Fit s	Sample s	RAM / VRAM	Hardware	CPU h, GPU h	Bytes
DOPE	7.25(13.5)	not recorded	920.7/ 1067	GPU host A RTX 4090; GPU host B, GPU host C RTX 3090	not recorded, 0.193	1,895.5(758–22,233)
Gaussian copula	not recorded	not recorded	not recorded	host not recorded	not recorded	5,464.5(892–833,439)
Chow–Liu, study	not recorded	not recorded	not recorded	host not recorded	not recorded	29,865(3,348–2,035,888)
Indep. marginals	not recorded	not recorded	not recorded	host not recorded	not recorded	2,549.5(862–36,672)
CTGAN	213.5(268.1)	186.7(197.0)	1557.8/ 1101	GPU host A RTX 4090	not recorded, 2.52	741,834(528,075–1,428,695)
TVAE	214.2(253.2)	186.1(196.6)	1252.9/ 1069	GPU host A RTX 4090	not recorded, 2.45	317,141.5(157,578–741,659)
Forest-Flow	185.9(380.6)	342.4(511.7)	6112.0/ 3181	GPU host A RTX 4090	not recorded, 2.44	325,848,913.5(115,202,253–1,085,155,062)
ARF	not recorded	not recorded	not recorded	GPU host B, GPU not recorded	not recorded	1,627,718.5(21,658–34,331,533)
TabSyn	not measured	not measured	not measured	not measured	not measured	not measured
TabDDPM	not measured	not measured	not measured	not measured	not measured	not measured

TABLE IV
STRONGEST TUNED BASELINE AT $4n$. W/T/L COUNTS LINEAGES. HOLM STAYS INSIDE THE BLOCK.

Block	Auditor	Baseline	n	Difference	W/T/L	Holm p
Density	CatBoost	Gaussian copula	97	0.224 [0.149, 0.290]	86/0/11	1.35×10^{-13}
Density	Linear	Gaussian copula	92	0.00366 [-0.00438, 0.012]	49/0/43	0.373
Density	MLP	Gaussian copula	75	0.185 [0.128, 0.228]	65/0/10	4.96×10^{-11}
Neural	CatBoost	TVAE	21	0.236 [0.087, 0.431]	19/0/2	1.57×10^{-4}
Neural	Linear	TVAE	21	0.263 [0.166, 0.410]	20/0/1	1.57×10^{-4}
Neural	MLP	TVAE	19	0.284 [0.153, 0.734]	17/0/2	1.57×10^{-4}
Forest	CatBoost	Forest-Flow	6	-0.027 [-0.308, 0.00804]	1/0/5	0.375
Forest	Linear	Forest-Flow	5	0.012 [-0.260, 0.302]	3/0/2	0.812
Forest	MLP	Forest-Flow	4	-0.032 [-0.227, -0.010]	0/0/4	0.375

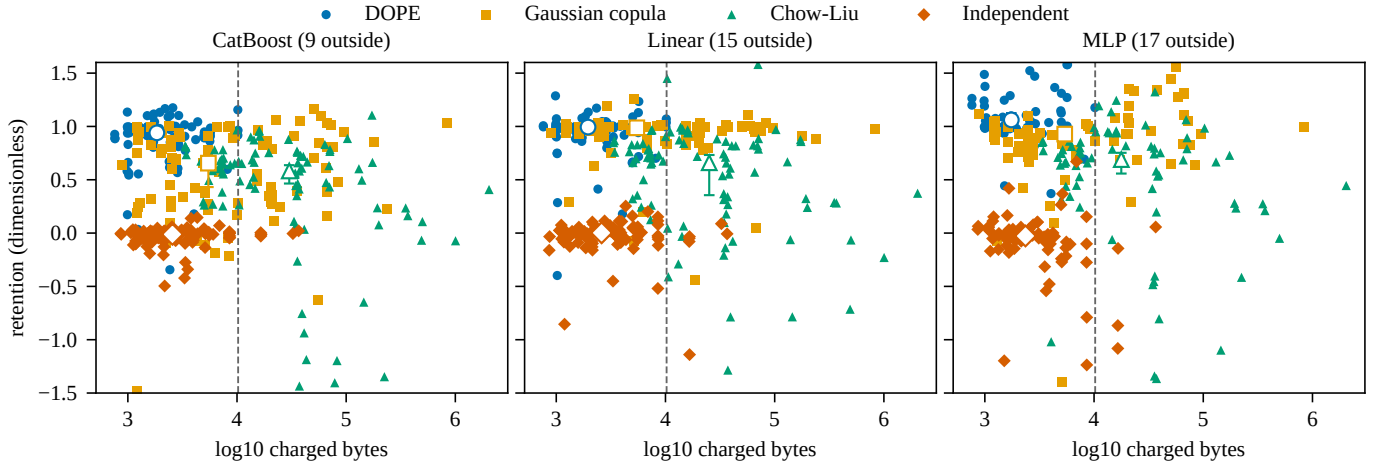


Fig. 2. Density-block retention at $4n$ against charged bytes. Three auditors. The dashed line is the 10,240-byte cap.

OriginalofMusic at 22,233 bytes and 588_fri_c4_1000_100 at 19,019 bytes. 678_visualizing_environmental has a fit whose CatBoost group is not informative, so the CatBoost median uses 97 lineages while the byte count uses 100. Independent marginals pass the cap more often than the study tree and still sit near retention zero. Smallness alone is not the result. The displayed byte range runs from 758 to 22,233.

On the density cells, the displayed profile has median classifier two-sample AUC 0.517 (interval 0.504 to 0.532, $n = 98$), mean marginal Kolmogorov–Smirnov distance 0.114 (0.103 to 0.135), and pairwise correlation fidelity 0.953 (0.945 to 0.961). A two-sample AUC near one half means a linear-time classifier has trouble telling the synthetic rows from the real validation rows. Table VIII adds alpha-precision, beta-

recall, DCR share, and DOMIAS AUC. The lower block is the neural lineages only. Holm families are 12 and 8, separate from retention. Higher DCR share and DOMIAS AUC sit nearer the fit view and are not a quality win. DOMIAS stays near one half. The density DCR contrast against independent marginals clears Holm in that worse direction. ARF awaits shared validation sampling. The production privacy-attack panel remains not measured. This is not differential privacy or HIPAA de-identification.

B. Neural block

Table IX is not pooled with Table V. On the 21 CatBoost lineages, DOPE’s median is 0.953, CTGAN’s is -0.081 , and

TABLE V
DENSITY BLOCK AT $4n$. W/T/L COUNTS LINEAGES.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	Gaussian copula	97	0.940 [0.914, 0.976]	0.656 [0.431, 0.803]	0.224 [0.149, 0.290]	86/0/11	1.35×10^{-13}
CatBoost	Chow-Liu, study	97	0.940 [0.914, 0.976]	0.586 [0.465, 0.637]	0.337 [0.282, 0.451]	94/0/3	4.85×10^{-16}
CatBoost	Indep. marginals, study	97	0.940 [0.914, 0.976]	-0.012 [-0.020, -0.00649]	0.956 [0.918, 0.980]	96/0/1	1.17×10^{-16}
Linear	Gaussian copula	92	0.994 [0.981, 0.996]	0.989 [0.968, 0.994]	0.00366 [-0.00438, 0.012]	49/0/43	0.373
Linear	Chow-Liu, study	92	0.994 [0.981, 0.996]	0.661 [0.356, 0.738]	0.340 [0.244, 0.425]	86/0/6	7.12×10^{-15}
Linear	Indep. marginals, study	92	0.994 [0.981, 0.996]	0.00206 [-0.023, 0.00583]	0.988 [0.975, 1.001]	92/0/0	5.70×10^{-16}
MLP	Gaussian copula	75	1.061 [1.024, 1.125]	0.928 [0.865, 0.953]	0.185 [0.128, 0.228]	65/0/10	4.96×10^{-11}
MLP	Chow-Liu, study	75	1.061 [1.024, 1.125]	0.692 [0.582, 0.754]	0.530 [0.350, 0.651]	74/0/1	2.11×10^{-13}
MLP	Indep. marginals, study	75	1.061 [1.024, 1.125]	-0.028 [-0.047, -0.013]	1.104 [1.047, 1.179]	75/0/0	2.11×10^{-13}

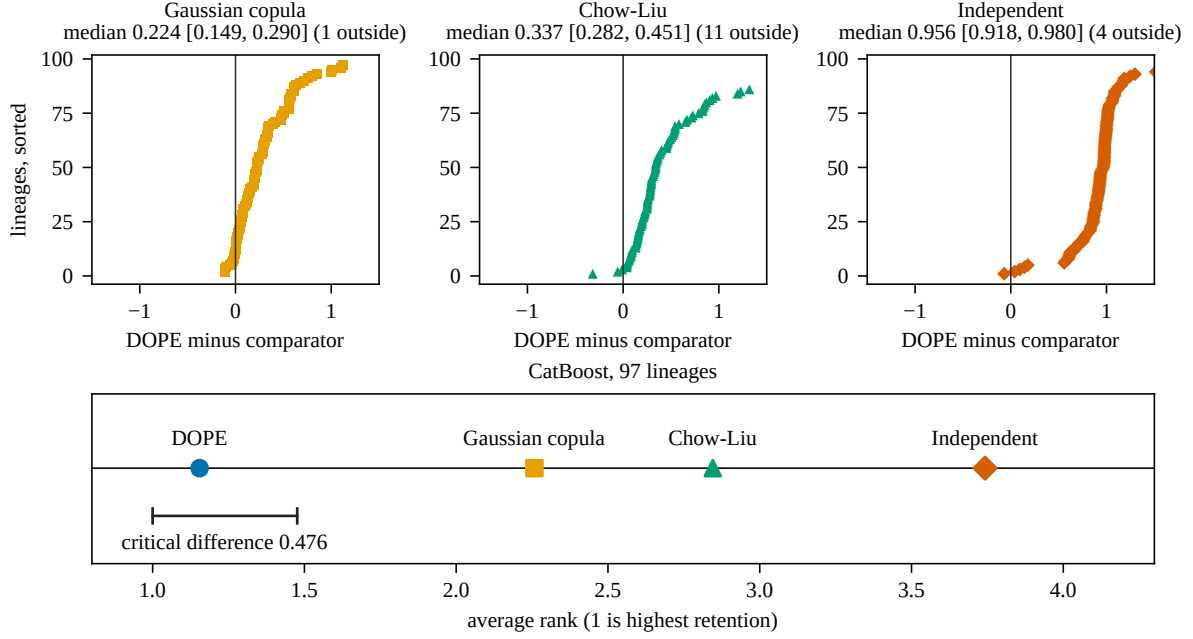


Fig. 3. CatBoost paired differences, sorted, and the Nemenyi scale. Rank 1 is the higher retention. The bar length is the critical difference.

TABLE VI
RAW AUDITOR MEDIANS ON THE INFORMATIVE LINEAGES.

Auditor	n	RMSE real	RMSE synth.	R^2 real	R^2 synth.
CatBoost	97	0.053	0.081	0.881	0.792
Linear	92	0.105	0.112	0.585	0.551
MLP	75	0.098	0.083	0.666	0.785

TABLE VII
CHARGED BYTES AT $4n$. FOREST-FLOW IS NOT POOLED.

Method	Median bytes	Within cap
DOPE reference	1,895.5	98/100
Gaussian copula	5,464.5	62/100
Chow-Liu, study	29,865	31/100
Indep. marginals, study	2,549.5	94/100
CTGAN	741,834	0/22
TVAE	317,141.5	0/22
Forest-Flow	325,848,913.5	0/6

TVAE's is 0.669. TVAE is the stronger neural baseline. Wins, ties, and losses against TVAE are 19/0/2, 20/0/1, and 17/0/2. The median favors DOPE, and DOPE still loses lineages. Every successful CTGAN and TVAE artifact in this byte count is over

the cap (0/22 and 0/22). Scheduling cutoffs are unfinished work, not method failures.

C. Forest-Flow

Table X is a 6-lineage confirmation, not pooled with the density or neural blocks. The Forest-Flow CatBoost median is 1.003 (interval 0.926 to 1.026) and the paired DOPE median is 0.940. Wins, ties, and losses are 1/0/5 under CatBoost, 3/0/2 under the linear auditor, and 0/0/4 under the MLP. DOPE is lower on every paired MLP lineage. The Holm tests are not significant at this count, so this is not a certified win for either method. Charged Forest-Flow artifacts run from 115,202,253 to 1,085,155,062 bytes (median 325,848,913.5) against a DOPE median of 2,067 on the same lineages. MFS-v2 on this block is null.

D. MFS-v2 column

Table XI is the scalar required by (3). No method has a complete component vector and a fully passed hard-gate map, so every entry is null. Copying a retention number into this

TABLE VIII
GROUPED-VALIDATION FIDELITY AND EMPIRICAL PRIVACY. NOT DP OR HIPAA.

Method	α -prec.	β -recall	DCR share	DOMIAS
<i>Median [interval] (lineages).</i>				
DOPE	0.905 [0.889, 0.918] (98)	0.437 [0.398, 0.473] (98)	0.811 [0.806, 0.814] (98)	0.508 [0.498, 0.520] (94)
Gaussian	0.812 [0.786, 0.832] (100)	0.430 [0.378, 0.453] (100)	0.806 [0.802, 0.811] (100)	0.513 [0.494, 0.527] (94)
Chow-Liu	0.858 [0.825, 0.904] (100)	0.359 [0.312, 0.400] (100)	0.810 [0.806, 0.816] (100)	0.512 [0.495, 0.522] (94)
Indep.	0.885 [0.868, 0.897] (100)	0.284 [0.197, 0.347] (100)	0.801 [0.797, 0.806] (100)	0.507 [0.496, 0.521] (94)
DOPE	0.902 [0.882, 0.940] (22)	0.441 [0.381, 0.489] (22)	0.811 [0.802, 0.819] (22)	0.508 [0.480, 0.538] (22)
CTGAN	0.719 [0.520, 0.856] (22)	0.170 [0.142, 0.267] (22)	0.820 [0.800, 0.842] (22)	0.502 [0.463, 0.534] (22)
TVAE	0.850 [0.731, 0.896] (22)	0.356 [0.283, 0.401] (22)	0.825 [0.813, 0.843] (22)	0.527 [0.487, 0.563] (22)

Contrast	α -prec.	β -recall	DCR share	DOMIAS
<i>Line 2: Holm p; rank-biserial r; n. DOPE minus comparator.</i>				
	0.083 [0.045, 0.113]	0.024 [0.011, 0.039]	-0.0025 [-0.00625, 0.005]	0.00157 [-0.0069, 0.0074]
DOPE—Gaussian	1.86×10^{-10} ; 0.781; 98 0.019 [0.00782, 0.056]	0.394; 0.215; 98 0.048 [0.029, 0.072]	1.000; 0.000206; 98 -0.00125 [-0.00562, 0.00125]	1.000; -0.085; 94 0.000581 [-0.00802, 0.011]
DOPE—Chow-Liu	2.00×10^{-5} ; 0.551; 98 0.018 [0.0072, 0.028]	1.32×10^{-10} ; 0.789; 98 0.104 [0.068, 0.138]	0.806; -0.169; 98 0.00625 [0.0025, 0.011]	1.000; 0.00572; 94 0.00244 [-0.00595, 0.015]
DOPE—Indep.	5.47×10^{-5} ; 0.523; 98	9.71×10^{-15} ; 0.937; 98	0.005; 0.398; 98	1.000; 0.080; 94
DOPE—CTGAN	0.168 [0.054, 0.368] 0.004; 0.794; 22 0.071 [0.00713, 0.141]	0.219 [0.131, 0.256] 3.81×10^{-6} ; 1.000; 22 0.062 [0.00546, 0.135]	-0.00812 [-0.029, 0.015] 0.532; -0.277; 22 -0.025 [-0.030, -0.00641]	-0.00028 [-0.014, 0.031] 0.610; 0.130; 22 -0.021 [-0.036, -0.00972]
DOPE—TVAE	0.040; 0.644; 22	0.069; 0.573; 22	0.059; -0.613; 22	0.108; -0.510; 22

TABLE IX
NEURAL BLOCK AT $4n$, NOT POOLED. W/T/L COUNTS LINEAGES.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	CTGAN	21	0.953 [0.896, 0.990]	-0.081 [-0.202, 0.103]	0.881 [0.702, 1.150]	21/0/0	5.72×10^{-6}
CatBoost	TVAE	21	0.953 [0.896, 0.990]	0.669 [0.539, 0.787]	0.236 [0.087, 0.431]	19/0/2	1.57×10^{-4}
Linear	CTGAN	21	0.999 [0.996, 1.015]	-0.033 [-0.225, 0.080]	1.049 [0.916, 1.236]	21/0/0	5.72×10^{-6}
Linear	TVAE	21	0.999 [0.996, 1.015]	0.796 [0.605, 0.867]	0.263 [0.166, 0.410]	20/0/1	1.57×10^{-4}
MLP	CTGAN	19	1.026 [0.998, 1.263]	-0.149 [-0.471, 0.083]	1.079 [0.877, 2.030]	19/0/0	1.53×10^{-5}
MLP	TVAE	19	1.026 [0.998, 1.263]	0.753 [0.478, 0.852]	0.284 [0.153, 0.734]	17/0/2	1.57×10^{-4}

TABLE X
FOREST-FLOW CONFIRMATION AT $4n$, CATBOOST. NOT POOLED.

Lineage	DOPE bytes	DOPE	Forest-Flow bytes	Forest-Flow
1193_BNG_lowbwt	2,436	0.970	416,921,273	1.004
503_wind	3,846	0.996	637,056,689	1.016
582_fri_c1_500_25	5,670	0.909	1,085,155,062	0.883
596_fri_c2_250_5	1,698	0.807	234,776,554	0.968
feynman_I_12_1	1,032	0.991	123,784,681	1.001
strogatz_vdp1	984	0.581	115,202,253	1.036

TABLE XI
MFS-v2 AFTER NATIVE SELECTION. NULL MEANS NOT EMITTED.

Method	MFS-v2
DOPE reference	null
Gaussian copula	null
CTGAN	null
TVAE	null
Forest-Flow	null
ARF	null
TabDDPM	null
TabSyn	not measured
synthpop CART	null
Chow-Liu, study code	null
Indep. marginals, study code	null

column would publish a value the contract refuses to emit. PTF-v1 is null. Release-safe certification is null. The admission locks are not frozen. TabSyn is not measured. TabDDPM on this 100-lineage panel is not measured.

E. BeyondArena

BeyondArena

(2ecfe882ccfb814fc27c4de10a64ceefd5d7655c) is not one of the 100 PMLB lineages and is not pooled with them.

5 of 12 families fit inside the cap. 7 did not prepare (4 outside the projected range, 3 overlapping). The range was not raised. Table XII has no retention column. The supplement names all 142 families.

VI. ABLATIONS

features12_steps8192 is the same residual with four times as many AdamW steps. It is a budget check, not the displayed model. Where both budgets have a complete three-seed group, the CatBoost median paired difference is 0.034 (68 lineages). The supplement gives the linear and MLP differences and lists all 100 lineages. A blank cell is an unavailable fit or

TABLE XII
BEYONDARENA FAMILIES THAT PREPARED. NO RETENTION AUDITOR.

Family	Fit rows	2,048 B	8,192 B	Val 2,048	Val 8,192
airfoil	802	804	818	0.010	0.00448
garments	939	2,317	2,372	0.028	0.025
musk	3,738	9,273	9,350	0.146	0.090
telemonitoring	257	1,852	1,890	0.033	0.019
video	35,918	5,054	5,071	0.00247	0.00195

TABLE XIII
INFORMATIVENESS THRESHOLD AND AUDITOR COUNTS.

Threshold	DOPE	Gaussian	Chow-Liu	Indep.
0	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.001	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.01	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.05	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.1	0.942 (96)	0.668 (98)	0.584 (98)	-0.012 (98)

Threshold	CatBoost	Linear	MLP
0.001	97	93	75
0.01	97	92	75
0.05	97	90	73

an auditor that is not informative. It is not a zero. The scored fits did not keep a per-step series. The supplement plots the logged replay. It is not a substitute for the scored artifacts. The lower loss at 8,192 steps does not replace the displayed profile.

The primary informativeness threshold is 0.01 times the absolute null loss. Table XIII lists the CatBoost median by threshold and the informative counts. The middle row is the 0.01 rule behind the retention tables. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Fit-seed variance across seeds other than 11 is not measured.

VII. DISCUSSION

CTGAN and TVAE do not match the CatBoost retention, and their successful size-4 artifacts miss the cap. None of these comparisons is an MFS-v2 score, a PTF-v1 result, or a release-safe certification. A small Wilcoxon p -value does not flip that null. The 8,192-step budget moves the CatBoost median by 0.034 and is not the displayed model. BeyondArena fits 5 further families inside the cap and does not show retention. TabSyn, a 100-lineage TabDDPM panel, and an equal tuning budget are not measured. Fit-seed variance is not measured.

VIII. THREATS TO VALIDITY

a) Internal.: The fits use coresets of 72 to 800 rows, one fit seed, and a training-derived validation cut of the official training rows. The supplement records the draw. The official test was not an input to training, selection, or the numbers in this paper. The informativeness rule drops a negligible real-versus-null gap. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Incomplete groups are omitted, not zeroed. Chow-Liu and independent marginals are study code. CTGAN and TVAE use the lineages that finished. Unfinished scheduling cutoffs are not method failures.

b) External.: These fits do not establish behavior on the full tables used by the diffusion papers, on a five-seed campaign, or on the official test. Generalization past seed 1729 is not measured. Forest-Flow’s byte comparison uses 6 lineages and the hash-checked fit files. BeyondArena is a separate fit check. It is not a retention panel, and it is not pooled with the density block.

c) Construct.: Retention is the reported utility. It is not the MFS-v2 scalar, which stays null. The linear auditor does not separate DOPE from the Gaussian copula. Table VIII is not a membership-inference certification. Its DCR share and DOMIAS AUC are empirical. Empirical resemblance is not differential privacy and is not HIPAA de-identification.

d) Statistical.: The rank tests are descriptive, not the pre-registered production superiority test. Density, neural, Forest-Flow, and BeyondArena stay unpooled. A small retention p -value does not flip the null scalar.

IX. CONCLUSION

On 97 validation lineages, `features12_steps2048` has CatBoost retention 0.940 (interval 0.914 to 0.976) and 98/100 artifacts inside the cap. CatBoost separates it from the density comparators; the linear auditor does not separate it from the Gaussian copula. Forest-Flow is ahead on CatBoost (1/0/5) and the MLP (0/0/4). The certification scalar is null, official tests stay sealed, and unrun generators are not measured.

REFERENCES

- [1] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, “Modeling tabular data using conditional GAN,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019. [Online]. Available: <https://papers.nips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html>
- [2] N. Patki, R. Wedge, and K. Veeramachaneni, “The synthetic data vault,” in *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, 2016, pp. 399–410.
- [3] D. S. Watson, K. Blesch, J. Kapar, and M. N. Wright, “Adversarial random forests for density estimation and generative modeling,” in *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 206, 2023, pp. 5357–5375. [Online]. Available: <https://proceedings.mlr.press/v206/watson23a.html>
- [4] A. Kotelnikov, D. Baranchuk, I. Rubachev, and A. Babenko, “TabDDPM: Modelling tabular data with diffusion models,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 17 564–17 579. [Online]. Available: <https://proceedings.mlr.press/v202/kotelnikov23a.html>
- [5] B. Nowok, G. M. Raab, and C. Dibben, “synthpop: Bespoke creation of synthetic data in R,” *Journal of Statistical Software*, vol. 74, no. 11, pp. 1–26, 2016.
- [6] C. K. Chow and C. N. Liu, “Approximating discrete probability distributions with dependence trees,” *IEEE Transactions on Information Theory*, vol. 14, no. 3, pp. 462–467, May 1968.
- [7] N. Park, M. Mohammadi, K. Gorde, S. Jajodia, H. Park, and Y. Kim, “Data synthesis based on generative adversarial networks,” *Proceedings of the VLDB Endowment*, vol. 11, no. 10, pp. 1071–1083, 2018.
- [8] J. Kim, C. Lee, and N. Park, “STaSy: Score-based tabular data synthesis,” in *International Conference on Learning Representations*, 2023. [Online]. Available: https://openreview.net/forum?id=1mNssCWt_v
- [9] C. Lee, J. Kim, and N. Park, “CoDi: Co-evolving contrastive diffusion models for mixed-type tabular synthesis,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 18 940–18 956. [Online]. Available: <https://proceedings.mlr.press/v202/lee23i.html>

- [10] H. Zhang, J. Zhang, B. Srinivasan, Z. Shen, X. Qin, C. Faloutsos, H. Rangwala, and G. Karypis, "Mixed-type tabular data synthesis with score-based diffusion in latent space," in *International Conference on Learning Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=4Ay23yeuz0>
- [11] J. Shi, M. Xu, H. Hua, H. Zhang, S. Ermon, and J. Leskovec, "TabDiff: a mixed-type diffusion model for tabular data generation," in *International Conference on Learning Representations*, 2025. [Online]. Available: <https://openreview.net/forum?id=swvURjrt8z>
- [12] M. Mueller, K. Gruber, and D. Fok, "Continuous diffusion for mixed-type tabular data," in *International Conference on Learning Representations*, 2025. [Online]. Available: <https://openreview.net/forum?id=QPtoBpn4IZ>
- [13] A. Jolicoeur-Martineau, K. Fatras, and T. Kachman, "Generating and imputing tabular data via diffusion and flow-based gradient-boosted trees," in *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 238. PMLR, 2024, pp. 1288–1296. [Online]. Available: <https://proceedings.mlr.press/v238/jolicoeur-martineau24a.html>
- [14] D. T. Zagardo, "Geometry-aware tabular diffusion," in *Proceedings of the 43rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 306. PMLR, 2026, pp. 152 812–152 835. [Online]. Available: <https://proceedings.mlr.press/v306/zagardo26a.html>
- [15] V. Borisov, K. Seßler, T. Leemann, M. Pawelczyk, and G. Kasneci, "Language models are realistic tabular data generators," in *International Conference on Learning Representations*, 2023. [Online]. Available: <https://openreview.net/forum?id=cEygmQNOeI>
- [16] A. V. Solatorio and O. Dupriez, "REaLTabFormer: Generating realistic relational and tabular data using transformers," 2023.
- [17] J. Ma, A. Dankar, G. Stein, G. Yu, and A. Caterini, "TabPFGGen – tabular data generation with TabPFN," in *NeurIPS 2023 Second Table Representation Learning Workshop*, 2023. [Online]. Available: <https://openreview.net/forum?id=4MkkNsAEmO>
- [18] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Körfer, S. B. Hoo, R. T. Schirmmeister, and F. Hutter, "Accurate predictions on small data with a tabular foundation model," *Nature*, vol. 637, no. 8045, pp. 319–326, 2025.
- [19] Z. Zhao, A. Kunar, R. Birke, and L. Y. Chen, "CTAB-GAN: Effective table data synthesizing," in *Proceedings of The 13th Asian Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 157. PMLR, 2021, pp. 97–112. [Online]. Available: <https://proceedings.mlr.press/v157/zhao21a.html>
- [20] Z. Zhao, A. Kunar, R. Birke, H. Van der Scheer, and L. Y. Chen, "CTAB-GAN+: Enhancing tabular data synthesis," *Frontiers in Big Data*, vol. 6, 2024.
- [21] T. Liu, Z. Qian, J. Berrevoets, and M. van der Schaar, "GOGGLE: Generative modelling for tabular data by learning relational structure," in *International Conference on Learning Representations*, 2023. [Online]. Available: <https://openreview.net/forum?id=fPVRcJqspu>
- [22] V. Borisov, T. Leemann, K. Seßler, J. Haug, M. Pawelczyk, and G. Kasneci, "Deep neural networks and tabular data: A survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 6, pp. 7499–7519, 2024.
- [23] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [24] J. Zhang, G. Cormode, C. M. Procopiuc, D. Srivastava, and X. Xiao, "PrivBayes: Private data release via bayesian networks," in *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*, 2014, pp. 1423–1434.
- [25] R. McKenna, G. Miklau, and D. Sheldon, "Winning the NIST contest: A scalable and general approach to differentially private synthetic data," *Journal of Privacy and Confidentiality*, vol. 11, no. 3, 2021.
- [26] R. McKenna, B. Mullins, D. Sheldon, and G. Miklau, "AIM: An adaptive and iterative mechanism for differentially private synthetic data," *Proceedings of the VLDB Endowment*, vol. 15, no. 11, pp. 2599–2612, 2022.
- [27] K. Cai, X. Lei, J. Wei, and X. Xiao, "Data synthesis via differentially private markov random fields," *Proceedings of the VLDB Endowment*, vol. 14, no. 11, pp. 2190–2202, 2021.
- [28] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *2017 IEEE Symposium on Security and Privacy (SP)*, 2017, pp. 3–18.
- [29] J. Hayes, L. Melis, G. Danezis, and E. De Cristofaro, "LOGAN: Membership inference attacks against generative models," *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 1, pp. 133–152, 2018.
- [30] B. van Breugel, H. Sun, Z. Qian, and M. van der Schaar, "Membership inference attacks against synthetic data through overfitting detection," in *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 206. PMLR, 2023, pp. 3493–3514. [Online]. Available: <https://proceedings.mlr.press/v206/breugel23a.html>
- [31] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–487, 2014.
- [32] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 2016, pp. 308–318.
- [33] J. Snoke, G. M. Raab, B. Nowok, C. Dibben, and A. Slavkovic, "General and specific utility measures for synthetic data," *Journal of the Royal Statistical Society Series A: Statistics in Society*, vol. 181, no. 3, pp. 663–688, 2018.
- [34] G. M. Raab, B. Nowok, and C. Dibben, "Practical data synthesis for large samples," *Journal of Privacy and Confidentiality*, vol. 7, no. 3, pp. 67–97, 2018.
- [35] A. Alaa, B. Van Breugel, E. S. Saveliev, and M. van der Schaar, "How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 2022, pp. 290–306. [Online]. Available: <https://proceedings.mlr.press/v162/ala22a.html>
- [36] K. A. Zhang, K. Veeramachaneni, and C. Sala, "SDMetrics: Synthetic data metrics," Open-source library, MIT Data To AI Lab, 2020. [Online]. Available: <https://docs.sdv.dev/sdmetrics/>
- [37] Z. Qian, R. Davis, and M. van der Schaar, "Synthcity: A benchmark framework for diverse use cases of tabular synthetic data," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 3173–3188, datasets and Benchmarks Track. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/09723c9f291f6056fd1885081859c186-Abstract-Datasets_and_Benchmarks.html
- [38] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 507–520.
- [39] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," in *Advances in Neural Information Processing Systems*, vol. 31, 2018. [Online]. Available: <https://papers.nips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>
- [40] R. Shwartz-Ziv and A. Armon, "Tabular data: Deep learning is not all you need," *Information Fusion*, vol. 81, pp. 84–90, 2022.
- [41] D. McElfresh, S. Khandagale, J. Valverde, V. Prasad C, G. Ramakrishnan, M. Goldblum, and C. White, "When do neural nets outperform boosted trees on tabular data?" in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 76 336–76 369, datasets and Benchmarks Track. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/f06d5ebd4ff40b40dd97e30cee632123-Abstract-Datasets_and_Benchmarks.html
- [42] N. Erickson, L. Purucker, A. Tschalzev, D. Holzmüller, P. Mutalik Desai, D. Salinas, and F. Hutter, "TabArena: A living benchmark for machine learning on tabular data," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2026, pp. 17 285–17 350. [Online]. Available: <https://openreview.net/forum?id=jZqCqpCLdU>
- [43] R. S. Olson, W. La Cava, P. Orzechowski, R. J. Urbanowicz, and J. H. Moore, "PMLB: a large benchmark suite for machine learning evaluation and comparison," *BioData Mining*, vol. 10, no. 1, p. 36, 2017.
- [44] J. D. Romano, T. T. Le, W. La Cava, J. T. Gregg, D. J. Goldberg, P. Chakraborty, N. L. Ray, D. Himmelstein, W. Fu, and J. H. Moore, "PMLB v1.0: an open-source dataset collection for benchmarking machine learning methods," *Bioinformatics*, vol. 38, no. 3, pp. 878–880, 2022.
- [45] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, no. 1, pp. 1–30, 2006. [Online]. Available: <https://jmlr.org/papers/v7/demsar06a.html>
- [46] A. Benavoli, G. Corani, J. Demšar, and M. Zaffalon, "Time for a change: a tutorial for comparing multiple classifiers through bayesian analysis," *Journal of Machine Learning Research*, vol. 18, no. 77, pp. 1–36, 2017. [Online]. Available: <https://jmlr.org/papers/v18/benavoli17.html>